

Scope of Work

Domu Technical Operations Platform
MVP & Production Engineering

Domu Technology Inc.

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Demo environment — representative data only. Not for use with real customer information.

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1 Executive Summary

The challenge describes a Technical Operations workflow that requires substantial manual coordination across client performance monitoring, QA investigation, engineering requests, and compliance-related operations. As client and call volume grows, these workflows can become a bottleneck and create opportunities for standardization and automation.

This document scopes an MVP and production Technical Operations platform that automates high-value repetitive workflows, surfaces operational intelligence, and routes the right issues to the right people — while preserving human control over consequential decisions.

The MVP focuses on three core capabilities: portfolio performance monitoring, QA call investigation, and client-request-to-ticket conversion. Four additional workflows are scoped for production. The SOW defines what the MVP proves, what production engineering must build, and how the system handles security, AI, observability, testing, and deployment.

2 Problem Statement

A Technical Operations Lead managing ~7 active clients and multiple voicebots currently must:

- Manually check each client for performance degradation — answer rate drops, conversion drops, QA issue spikes.
- Manually listen to flagged calls, classify issues, assess severity, and formulate recommendations.
- Manually translate unstructured client requests into engineering-ready tickets.
- Manually verify whether proposed call attempts comply with configured calling-window rules.
- Investigate compliance escalations, voicebot behavior regressions, and new client onboarding — each requiring the same repeated evidence-gathering steps.

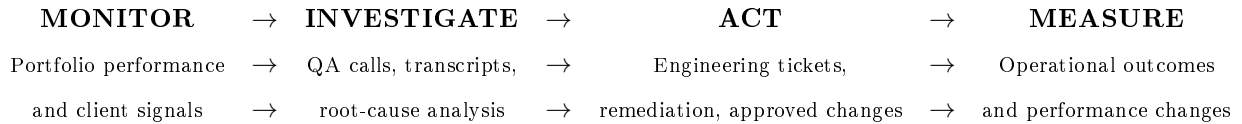
These workflows contain repetitive components that are strong candidates for standardization and automation, allowing senior operational staff to focus more of their time on investigation, judgment, and client outcomes. Inconsistency in these tasks introduces risk: missed performance degradation, inconsistent QA findings, and poorly scoped engineering work.

3 Proposed Solution

A purpose-built Technical Operations platform that:

1. Surfaces a single prioritized portfolio view of all client/voicebot performance.
2. Provides a structured QA investigation workflow with AI-assisted transcript classification and human-controlled dispositions.
3. Converts unstructured client requests into engineering-ready ticket drafts, surfacing open questions before work begins.
4. Deterministically validates calling-window compliance against versioned client/regional policies.
5. In production: supports structured voicebot flow creation, prompt diagnosis, and compliance investigation.

Deterministic logic handles rule-based tasks. AI assists with unstructured text. Humans retain final control over all consequential decisions.



The platform is designed around a closed operational loop: detect a problem, investigate evidence, take a controlled action, and measure the resulting outcome.

4 MVP Scope and Prioritization

The challenge asks for automation of *some* repetitive operations tasks. This MVP demonstrates three core workflows using representative seeded data. Remaining workflows are scoped for production.

#	Capability	Priority	MVP?
1	Portfolio Performance Monitoring	Core MVP	Yes — seeded data
2	QA / Voicebot Call Investigation	Core MVP	Yes — seeded data
3	Client Request → Engineering Ticket	Core MVP	Yes — seeded data
4	Calling-Window Validation	Supporting	Yes — demo rules
5	Client Script → Voicebot Flow	Extended	Production only
6	Voicebot Prompt Diagnosis	Extended	Production only
7	Compliance Concern Investigation	Extended	Production only

Table 1: Feature prioritization and MVP scope.

The MVP uses representative seeded data with no production credentials and no real customer information. The UI clearly identifies the demo environment.

Features 5–7 are intentionally excluded from the MVP implementation. They represent additional production capabilities identified from the broader Technical Operations workflow in the challenge and can be evaluated after the core MVP has been validated.

5 Functional Capabilities

5.1 Feature 1 — Portfolio Performance Monitoring

Objective. Answer: *"Across all active clients, where do I need to spend my attention right now?"* The portfolio view provides deterministic KPIs, per-client status, and a prioritized action list without requiring the operator to query each account manually.

MVP. Dashboard with seeded data showing portfolio-level and per-client KPIs, a client performance table (answer rate, payment conversion, QA issue count, trend, status), a prioritized action list, and client-detail drill-down. All calculations use standard arithmetic — no AI for metrics.

Metric	Definition
Answer Rate	Answered calls / total calls
Payment Conversion	Payments / answered calls
QA Issue Count	Open issues associated with the client
Client Status	Deterministic: Healthy / Needs Attention / Critical

Table 2: Core portfolio metrics (deterministic).

Production direction. Replace seeded data with authenticated integrations to call-event, outcome, QA, and client-configuration sources. Add freshness indicators, ingestion-failure visibility, configurable thresholds, client/account permissions, and KPI reconciliation.

Acceptance criteria.

1. KPIs computed correctly from seeded data; no hard-coded display values.
2. Division-by-zero handled safely (display "N/A" or equivalent).
3. At least one client clearly flagged in the demo data set.
4. Priority actions identify a specific problem, not a generic flag.
5. Demo environment label visible; no implication of real data.

5.2 Feature 2 — QA / Voicebot Call Investigation

Objective. Answer: *"What went wrong in this call, why, and what should I do next?"* Reduces manual effort required to locate, classify, and action problematic calls.

MVP. QA queue with flagged calls (filterable by client, severity, category, status), call-detail view with transcript, AI-assisted or seeded classification (category, severity, evidence, recommendation), human review actions (Accept / Reject / Escalate), and engineering-ticket handoff. Uses the challenge-supplied Hannah voicebot and call recordings.

Production direction. Real call/transcript integrations, automated QA sampling, model versioning, labeled evaluation data, regression evaluation before model/prompt changes, and full auditability per the global auditability requirements.

Human control. The system must not automatically modify voicebots, deploy configuration, or make compliance determinations. Human review is required before findings are actioned.

Acceptance criteria.

1. Queue loads; operator can filter by client, severity, and status.
2. Each finding has category, severity, evidence, and recommendation.
3. Evidence is traceable to a specific transcript segment.
4. Accept / Reject / Escalate update finding state correctly.
5. Engineering-ticket handoff populates all required fields.
6. Demo remains deterministic without a live external AI service.

5.3 Feature 3 — Client Request → Engineering Ticket

Objective. Convert unstructured client requests into engineering-ready work items, surfacing open questions before work begins rather than after.

MVP. Client request view, AI-assisted structured ticket generation (title, user story, acceptance criteria, dependencies, open questions, priority, observability considerations), editable draft, copy/export, and QA-issue-to-ticket handoff. Reference scenario: *"We'd like customers to be able to pay via Pix."*

Production direction. Engineering issue-tracker integration (create, link, sync status), full traceability from original request through implementation and measurement.

Human control. Generated output is always a draft. The operator must review and approve every field before the ticket is marked ready for engineering.

Acceptance criteria.

1. Structured ticket generated from representative request includes all required fields.
2. Open questions explicitly listed; known vs. unknown clearly distinguished.
3. All fields editable; draft status visible.
4. QA-issue context pre-populates relevant ticket fields.
5. Demo works without a live external AI service.

5.4 Feature 4 — Calling-Window Validation

Objective. Determine whether a proposed call is allowed under the configured client/region calling policy. Deterministic, explainable, and auditable — not a legal determination.

MVP. Operator selects client, region, and proposed date/time. System evaluates against seeded calling policy (timezone, allowed window, holidays) and returns: **ALLOWED** / **BLOCKED** / **HOLIDAY** / **REQUIRES REVIEW** with a human-readable explanation. Logic is standard application code — no AI.

Production direction. Versioned managed policy data, reliable timezone and holiday sources, auditability of decisions (client, region, timestamp, policy version, result), exception-handling workflow.

Acceptance criteria.

1. Correct result for in-window, out-of-window, and holiday cases.
2. Missing or ambiguous policy returns **REQUIRES REVIEW**, not a guessed result.
3. Every result includes a human-readable explanation and the policy used.
4. Evaluation uses target-region timezone, not operator's local time.

5.5 Feature 5 — Client Script → Structured Voicebot Flow (Extended)

Objective. Convert an unstructured client-provided script into a structured voicebot conversation flow (branches, conditions, draft prompt), ready for human review and controlled deployment.

Production direction. AI-assisted extraction of conversation branches and conditions, operator review and editing, versioning of generated flows, human approval gate before any production voicebot deployment, rollback to previous known-good version, regression evaluation for behavior-change validation. Autonomous production deployment is explicitly out of scope.

5.6 Feature 6 — Voicebot Prompt Diagnosis & Improvement (Extended)

Objective. Investigate a client report of poor voicebot behavior, identify the root cause, propose a targeted prompt/workflow change, and validate the change before staged production rollout.

Production direction. Call evidence review, observed-vs-expected behavior comparison, inspection of relevant prompt/workflow branch, AI-assisted change proposal, regression testing with representative scenarios, human approval, staged rollout, and performance measurement. Engineering handoff when the root cause is not prompt-related.

5.7 Feature 7 — Compliance Concern Investigation (Extended)

Objective. Investigate a client escalation about a voicebot statement or behavior that may violate a business or regulatory requirement. Produce a structured internal finding and a draft

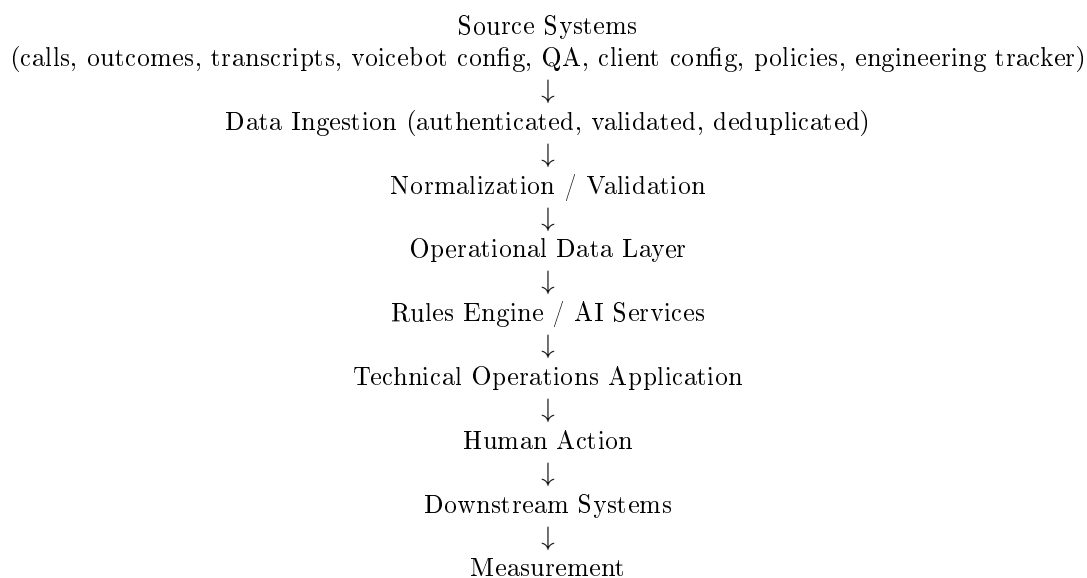
client response for human review.

Scope distinction. Calling-Window Validation asks *"Was this call permitted?"*. Compliance Concern Investigation asks *"Did something during this call potentially violate a requirement?"*

Production direction. Client escalation preserved verbatim, call identification, transcript evidence review, versioned policy/rule reference, observed-vs-expected comparison, AI-assisted investigation draft (advisory only), operator-assigned severity (Low / Medium / High / Critical), structured internal investigation record, client-response draft clearly marked "DRAFT — REQUIRES HUMAN REVIEW", human review and disposition (Confirm / Reject / Insufficient Evidence / Escalate), remediation tracking linked to QA or engineering work, full audit trail. This capability is not legal advice and does not replace compliance or legal professionals.

6 Production Architecture

6.1 Conceptual Data Flow



A shared operational data layer serves all seven workflows, eliminating duplicate pipelines and business logic per feature.

6.2 Core Data Model

6.3 Ingestion and Data Quality

Production ingestion must support: authenticated access, schema and input validation, duplicate detection and idempotency, retry/dead-letter handling, operator-visible failure states, source identifier preservation, and data freshness indicators (e.g., "Last updated 4 min ago" / "Ingestion delayed 47 min").

Data-quality failures must be visible. The platform must not silently compute misleading KPIs from incomplete data.

6.4 Versioning and Source-of-Truth

Voicebot configuration, calling policies, prompt templates, QA taxonomy, and AI/model configurations must be versioned so the system can answer: *"Which configuration and policy were active when this call occurred?"*

Entity	Key Attributes
Client	id, name, status, configuration, supported regions
Voicebot	id, clientId, name, active version, configuration reference, status
Call	id, clientId, voicebotId, timestamp, duration, outcome, answer status, transcript reference, config version
QA Issue	id, callId, clientId, category, severity, evidence, recommendation, review state, reviewer
Client Request	id, clientId, original request, structured requirement, status
Engineering Ticket	id, clientId, source request/issue, title, acceptance criteria, priority, status, external reference
Calling Policy	id, clientId, region, timezone, allowed window, holiday config, effective date, version
Voicebot Config Version	id, voicebotId, version, prompt reference, approvedBy, status, effective date

Table 3: Core operational data entities.

The platform maintains references to source-of-truth systems (call platform, payment system, engineering tracker) rather than replacing them.

7 Data & Integrations

The MVP uses seeded representative data. Production integrations connect to source systems via authenticated interfaces. The SOW does not assume specific technologies or vendors; implementation choices will be aligned with the team’s existing infrastructure standards and engineering practices.

Source Category	Consumed By
Call events and outcomes	Features 1, 2, 4, 6, 7
Transcripts	Features 2, 6, 7
Voicebot configuration	Features 2, 5, 6
QA results	Features 1, 2
Client configuration	All features
Calling policies / calendar	Feature 4
Engineering tracker	Feature 3
Payment / outcome systems	Feature 1

Table 4: Source-system consumption by feature.

8 AI-Assisted Automation & Human Oversight

8.1 Task Allocation

8.2 AI Governance

- **Traceability.** Every AI-generated finding is linked to its source call, transcript segment, and model/prompt version.
- **Evidence first.** AI output identifies evidence; it does not substitute for it.
- **Evaluation dataset.** Production maintains a labeled dataset for regression evaluation before model or prompt changes.
- **Human override.** Operators can accept, reject, or escalate any AI-generated finding.

Approach	Tasks
Deterministic (no AI)	KPI calculations, calling-window evaluation, holiday checks, status/priority rules, field validation
AI-assisted	Transcript classification and summarization, requirement extraction from client requests, ticket drafting, prompt change proposals, compliance investigation assistance
Human-controlled only	Compliance decisions, production prompt/config deployments, high-risk QA finding dispositions, final client-facing responses

Table 5: Task allocation: deterministic, AI-assisted, and human-controlled.

- **Prompt versioning.** Prompt templates are versioned alongside application code. Changes require review and staging before production rollout.
- **AI data boundary.** Only data required for the specific AI task is sent to an external AI service. PII and sensitive content are redacted where technically feasible. AI inputs, outputs, and model versions are logged.

The MVP may implement AI-assisted features using deterministic templates, controlled model output, or mock AI services, provided the demo is reproducible without an unreliable live external dependency.

9 Security & Compliance

The following requirements apply to all production capabilities. Individual features note additional feature-specific considerations where relevant.

- **Authentication & access.** All production endpoints require authenticated access. SSO integration where appropriate to the deployment environment.
- **RBAC.** Role-based access control with least-privilege defaults. Operational roles separated from administrative roles.
- **Tenant/client isolation.** Strict data isolation between client accounts. No cross-client data access.
- **PII minimization.** Collect and process only the data required. Avoid persisting sensitive customer data unnecessarily.
- **Transcript and audio access.** Access to call transcripts and recordings restricted by role and client scope.
- **Encryption.** Data encrypted in transit and at rest.
- **Audit logging.** All significant operational actions logged with user identity, timestamp, and record identifier. Human actions distinguished from system-generated actions.
- **Retention policies.** Data retained only as long as operationally and legally required.
- **AI input boundaries.** AI services receive only the minimum data required. Inputs logged alongside AI outputs and model versions.
- **Production/development separation.** Development credentials must not provide production access.

The MVP does not require full production security infrastructure. The MVP must not use real customer data.

10 Observability & Reliability

10.1 System Observability

- API latency and error rates
- Ingestion pipeline health, lag, and failure states
- Job/queue failures
- Data freshness — last successful refresh visible to operators
- Deployment health and version in flight

10.2 Operational Observability

- Call volume, answer rate, payment conversion per client
- QA issue rate and open-issue backlog
- Calling-window violations
- AI finding acceptance/rejection rates (quality proxy)

Degraded states must be visible to operators before they affect decisions (e.g., "Call data delayed — KPIs may be stale"). Alerts should be defined for critical ingestion failures and significant metric regressions.

11 Testing & Quality Assurance

- **Unit tests.** All deterministic logic: KPI calculations, calling-window evaluation, policy-precedence rules, field-validation logic.
- **Integration tests.** Data ingestion, normalization, entity relationships, API contracts.
- **End-to-end tests.** Core MVP workflows from data input to operator-facing output.
- **Golden dataset.** Labeled representative calls and requests for regression testing of AI-assisted classification and ticket generation.
- **AI evaluation.** Before any model or prompt change, regression evaluation against the golden dataset. Agreed accuracy thresholds define a passing gate.
- **Security testing.** Tenant isolation, access-control boundaries, input validation.
- **Acceptance testing.** Per-feature acceptance criteria (Sections 5.1–5.4) verified against the demo environment.

12 Deployment / CI/CD / Environments

12.1 MVP Deployment

The MVP will be deployed to an accessible URL for evaluation and will use representative seeded data only. It will not contain production credentials or real customer information.

12.2 Production Environments

- **Development** — feature work, local testing.
- **Staging** — integration testing, production-like validation, AI/prompt regression.

- **Production** — real client operations, approved workflows only.

Configuration, credentials, and data access are isolated between environments.

12.3 CI/CD Pipeline

Automated pipeline: source checkout → dependency install → lint / static analysis → unit tests → integration tests → build → security/dependency scan → staging deployment → staging validation → controlled production promotion. Releases must not require undocumented manual steps.

12.4 Quality Gates, Rollback, and Configuration

Failed tests, failed builds, or critical security issues block production promotion. Every production change has a defined rollback path. Secrets are managed through a secure mechanism (not source control). Environment-specific configuration is separated from application code. Data migrations are versioned and validated. AI/prompt changes are versioned, regression-evaluated, and staged before broad rollout.

Implementation technology (language, framework, cloud provider, CI/CD vendor, database, IaC tool) is not mandated by this SOW; decisions are made during engineering discovery aligned with Domu’s existing standards.

13 Rollout Plan

Phase	Name	Scope
0	MVP / Proof of Concept	Demo environment, seeded data, Features 1–4, interview evaluation.
1	Single-client Pilot	Features 1–4 on production-connected data for one client. Success criteria defined before Phase 2.
2	Multi-client Rollout	Expand to additional clients after pilot success. Evaluate and introduce Features 5–7 based on pilot findings, business priority, and engineering readiness.
3	Scale & Automation	Full portfolio, extended automation, performance tuning.

Table 6: Phased rollout.

Each phase uses feature flags for controlled exposure and retains rollback capability. Success criteria are defined and measured before promotion to the next phase.

14 Risks, Assumptions & Dependencies

14.1 Risks

14.2 Assumptions

- Production source systems exist or can be instrumented to expose required operational data via authenticated interfaces.
- Engineering can provide integration support for ingestion and downstream system connections.
- Client calling policies can be expressed as structured, versioned configuration data.

Risk	Mitigation
AI hallucination or incorrect classification	Evidence traceability + human review + golden-dataset regression evaluation
Bad or incomplete source data	Schema validation + reconciliation + freshness monitoring + visible failure states
Prompt/model regression	Regression suite + staged rollout + rollback
Compliance false positive or false negative	Human review required; AI advisory only; policy version referenced
Client-data exposure	RBAC + tenant isolation + PII minimization + audit logging
Operational overload on operators	Prioritization logic + automation of repetitive tasks + alerting
Scope creep into extended features	Explicit MVP/production distinction; phase gates before new capabilities

Table 7: Risk register.

- Compliance policy/rules are defined and maintained by the responsible business/compliance team.
- Production architecture will align with Domu’s existing technology standards and engineering practices.

15 Non-Goals

- Replacing the voicebot runtime or call platform
- Replacing Domu’s source-of-truth financial or transactional systems
- Replacing the engineering issue tracker
- Autonomous regulatory or legal determinations
- Autonomous production prompt or configuration deployment (human approval always required)
- Building a general-purpose data warehouse or BI platform
- Fully autonomous Technical Operations (human oversight is a core design principle)
- Prescribing Domu’s internal technology choices (cloud, language, vendor)

16 Definition of Done

Production readiness requires all of the following:

Area	Criteria
Functional	Core workflows (Features 1–4) operate correctly on production-connected data; extended features pass staging acceptance tests before promotion.
Data	KPIs reconcile with source systems; freshness visible; ingestion failures operator-visible.
AI	AI evaluation meets agreed accuracy thresholds on golden dataset; high-risk decisions require human review; model/prompt versions tracked.
Security	RBAC, tenant isolation, audit logging, and access controls verified; security review completed.
Observability	System and operational metrics instrumented; alerts configured for critical failures.
Deployment	CI/CD pipeline operational; quality gates enforced; rollback tested.
Rollout	Phase 1 pilot success criteria met and documented before Phase 2 promotion.

Table 8: Production definition of done.